

Lab experiment 1 :

```
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lexical.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
enter the string : a=b+c*10
identifiers : a b c
constants : 10
operators :
```

Lab experiment 2 :

```
C:\Windows\System32\cmd.e  X  +  v

Directory of C:\compiler design

28-01-2026  09:48      <DIR>          .
28-01-2026  09:48             63,547 a.exe
25-01-2026  18:37             311 labexp1.l
25-01-2026  18:43             288 labexp2.l
25-01-2026  18:59             302 labexp3.l
25-01-2026  19:07             329 labexp4.l
25-01-2026  19:11             271 labexp5.l
25-01-2026  19:17             552 labexp6 c program.txt
25-01-2026  19:19             611 labexp7 c program.txt
25-01-2026  19:24             1,376 labexp8 c program.txt
25-01-2026  19:39             269 labexp9.l
28-01-2026  09:48            37,714 lex.yy.c
08-01-2026  14:56              74 path's.txt
               12 File(s)          105,644 bytes
               1 Dir(s)  200,913,829,888 bytes free

C:\compiler design>set path=C:\Program Files (x86)\GnuWin32\bin

C:\compiler design>flex labexp2.l

C:\compiler design>set path=C:\Program Files\CodeBlocks\MinGW\bin

C:\compiler design>gcc lex.yy.c

C:\compiler design>a.exe
Enter a command : // this is a command
Single-line comment
```

Lab Experiment 3 :

```
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe

Enter any operator:+

Addition
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>
```

Lab Experiment 4 :

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc space.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
Enter text (up to 100 characters, use ~ to end):
void main()
{
int a;
int b;
a = b + c;
c = d * e;
}

^Z
Total number of words: 18
Total number of lines: 9
Total number of characters: 34

```

Lab Experiment 5 :

```

C:\Windows\System32\cmd.e  x  +  v
25-01-2026  19:11          271 labexp5.l
25-01-2026  19:17          552 labexp6 c program.txt
25-01-2026  19:19          611 labexp7 c program.txt
25-01-2026  19:24        1,376 labexp8 c program.txt
25-01-2026  19:39          269 labexp9.l
28-01-2026  09:54        37,521 lex.yy.c
08-01-2026  14:56           74 path's.txt
          12 File(s)          105,487 bytes
           1 Dir(s) 200,909,398,016 bytes free

C:\compiler design>set path=C:\Program Files (x86)\GnuWin32\bin

C:\compiler design>flex labexp5.l

C:\compiler design>set path=C:\Program Files\CodeBlocks\MinGW\bin

C:\compiler design>gcc lex.yy.c

C:\compiler design>a.exe
Enter an identifier:
8
Invalid Identifier

+
Invalid Identifier

a
Valid Identifier

```

Lab experiment 6 :

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lr.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
Enter Number of Production : 2
Enter the grammar as E->E-A :
S->(L)|a
L->L,S|S

GRAMMAR : : : S->(L)|a is not left recursive.

GRAMMAR : : : L->L,S|S is left recursive.
Grammar without left recursion:
L->SL'
L'->,L'|E

```

Lab experiment 7 :

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lf.c
lf.c: In function 'main':
lf.c:11:5: warning: call to 'gets' declared with attribute warning: Using ge
ts() is always unsafe - use fgets() instead [-Wattribute-warning]
   11 |     gets(gram);
      |     ^~~~~~

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
Enter Production: S->iEtS|iEtSeS|a

S->iEtSX
X->|eS|a

```

Lab experiment 8 :

main.c

Share

Run

Output

```

1 #include <stdio.h>
2 #include <string.h>
3
4 struct symbol {
5     char name[20];
6     char type[20];
7 };
8
9 int main() {
10     struct symbol table[10];
11     int n = 0, choice, i;
12     char key[20];
13
14     while (1) {
15         printf("\n1.Insert\n2.Search\n3.Display\n4.Exit\n");
16         printf("Enter your choice: ");
17         scanf("%d", &choice);
18
19         switch (choice) {
20
21             case 1:
22                 printf("Enter symbol name: ");
23                 scanf("%s", table[n].name);
24                 printf("Enter symbol type: ");
25                 scanf("%s", table[n].type);
26                 n++;
27                 break;
28

```

```

Enter symbol name: a
Enter symbol type: int

1.Insert
2.Search
3.Display
4.Exit
Enter your choice: 1
Enter symbol name: b
Enter symbol type: int

1.Insert
2.Search
3.Display
4.Exit
Enter your choice: 3

Symbol Table:
Name    Type
a      int
b      int
2      int

1.Insert
2.Search
3.Display
4.Exit
Enter your choice:

```

Lab experiment 9 :

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc grammar.c
grammar.c: In function 'main':
grammar.c:9:2: warning: call to 'gets' declared with attribute warning: Using gets() is always unsafe - use fgets() instead [-Wattribute-warning]
   9 |     gets(string);
     |     ^~~~~~

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
The grammar is: S->aS, S->Sb, S->ab
Enter the string to be checked:
abb
String accepted
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>

```

Lab experiment 10 :

main.c

Share

Run

Output

```

34 // Create new production
35 j = 0;
36 for (i = pos; part1[i] != '\0'; i++)
37     newGram[j++] = part1[i];
38
39     newGram[j++] = '|';
40
41     for (i = pos; part2[i] != '\0'; i++)
42         newGram[j++] = part2[i];
43
44     newGram[j] = '\0';
45
46 // Output
47 printf("\nAfter Left Factoring:");
48 printf("\nS -> %s", modifiedGram);
49 printf("\nX -> %s\n", newGram);
50
51     return 0;
52 }
53

```

```

Enter Production (example: ab|ac): S->iEtS|iEtSeS|a

After Left Factoring:
S -> X
X -> S->iEtS|iEtSeS|a

=== Code Execution Successful ===

```

Lab experiment 11 :

```
main.c
84 printf("no such symbol");
85 else
86 {
87 printf("label is found \n");
88 printf("enter the address");
89 scanf("%d",&ad);
90 for(i=0;i<cnt;i++)
91 {
92 if(strcmp(sy[i].label,lab)==0)
93 sy[i].addr=ad;
94 }
95 }
96 }
97 void display()
98 {
99 int i;
100 for(i=0;i<cnt;i++)
101 printf("%s\t%d\n",sy[i].label,sy[i].addr);
102 }
103
```

Output

```
enter the address200
1.insert
2.display
3.search
4.modify
5.exit
2
a 100
b 200
1.insert
2.display
3.search
4.modify
5.exit
```

Lab experiment 12 :

```
main.c
14 char lhs[10], a[10], b[10], c[10];
15 char op1, op2;
16
17 printf("Enter expression (example: x = a + b - c):\n");
18 scanf("%s = %s %c %s %c %s", lhs, a, &op1, b, &op2, c);
19
20 printf("\nTHREE ADDRESS CODE\n");
21
22 // First TAC
23 printf("t1 = %s %c %s\n", a, op1, b);
24
25 // Second TAC
26 printf("t2 = t1 %c %s\n", op2, c);
27
28 // Final assignment
29 printf("%s = t2\n", lhs);
30
31 return 0;
32 }
33
```

Output

```
Enter expression (example: x = a + b - c):
x=a-b
x=a+b-c

THREE ADDRESS CODE
t1 = a - b
t2 = t1 - c
x=a-b = t2

=== Code Execution Successful ===
```

Lab experiment 13 :

Total number of characters : 34

Lab Experiment 14 :

ProgramizC Online Compiler

main.c

Share

Run

```
55     stack[st_ptr - 1] == '*' ||
56     stack[st_ptr - 1] == '/' ) &&
57     stack[st_ptr - 2] == 'E')
58 {
59     stack[st_ptr - 2] = 'E';
60     stack[st_ptr - 1] = '\0';
61     st_ptr = st_ptr - 1;
62
63     printf("\n%s\t\t%s\t\tREDUCE E->E*cE",
64           stack, input, stack[st_ptr]);
65 }
66
67 // Accept conditional
68 if (strcmp(stack, "E") == 0 && ip_ptr == len)
69 {
70     printf("\n%s\t\t%s\t\tACCEPT", stack, input);
71     exit(0);
72 }
73 }
74
```

Output

SHIFT REDUCE PARSER

GRAMMAR

E -> E+E | E*E | E/E | a | b

Enter the input string: a+b

STACK	INPUT	ACTION
\$	a+b\$	--
\$a	+b\$	SHIFT
\$E	+b\$	REDUCE E->E
\$E+	b\$	SHIFT
\$E+b	\$	SHIFT
\$E+E	\$	REDUCE E->E
\$E	\$	REDUCE E->E E
\$E	\$	ACCEPT

=== Code Execution Successful ===

Lab experiment 15 :

```

main.c
120     printf("\n");
121     dispstack();
122     printf("\t");
123     dispinput();
124     printf("\tREDUCE E->%s", lasthandle);
125 }
126 else
127     break;
128 }
129 }
130
131 if (strcmp(stack, "$E$") == 0)
132     printf("\nAccepted\n");
133 else
134     printf("\nNot Accepted\n");
135
136 free(input);
137 return 0;
138 }
139

```

Output

Enter the string: i*(i+i)*i

STACK	INPUT	ACTION
\$i	*(i+i)*i\$	SHIFT
\$E	*(i+i)*i\$	REDUCE E->i
\$E*	(i+i)*i\$	SHIFT
\$E*(	i+i)*i\$	SHIFT
\$E*(i	+)i*\$	SHIFT
\$E*(E	+)i*\$	REDUCE E->i
\$E*(E+	i)*i\$	SHIFT
\$E*(E+i	)*i\$	SHIFT
\$E*(E+E	)*i\$	REDUCE E->i
\$E*(E+E)	*i\$	SHIFT
\$E*(E+E)*	i\$	SHIFT
\$E*(E+E)*i	\$	SHIFT
\$E*(E+E)*E	\$	REDUCE E->i
\$E*(E+E)*E\$		SHIFT
Not Accepted		

Lab experiment 16 :

```

operable program or batch file.

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex lines.l
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
#define PI 3.14
1  #define PI 3.14
#include<stdio.h>
2  #include<stdio.h>
#include<conio.h>
3  #include<conio.h>
void main()
4  void main()
{
5  {
int a,b,c = 30;
6  int a,b,c = 30;
printf("hello");
7  printf("hello");
}
8  }

```

Lab experiment 18 :

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex captial.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
Enter String :
MAHESH babu
MAHESH is a capital word

```

Lab experiment 19 :

```

10 File(s)      113,126 bytes
 2 Dir(s)  786,789,511,168 bytes free

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex mobile.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe

enter the mobile number:9392753160

mobile number valid

|

```

Lab experiment 20 :

main.c	Output
5	F + F
6	F * F
7	F (T
8	F) F
9	F i T
10	F \$ F
11	T + F
12	T * T
13	T (T
14	T) F
15	T i T
16	T \$ F
17	
18	FIRST SETS
19	
20	FIRST(E) = { + * (i }
21	FIRST(F) = { (i }
22	FIRST(T) = { * (i }
23	

Lab experiment 21 :

```
C:\Windows\System32\cmd.e  ×  +  v
28-01-2026  10:42          381 labexp21.l
25-01-2026  18:59          302 labexp3.l
25-01-2026  19:07          329 labexp4.l
25-01-2026  19:11          271 labexp5.l
25-01-2026  19:17          552 labexp6 c program.txt
25-01-2026  19:19          611 labexp7 c program.txt
25-01-2026  19:24        1,376 labexp8 c program.txt
28-01-2026  10:14        1,006 labexp9.l
28-01-2026  10:09       38,229 lex.yy.c
08-01-2026  14:56           74 path's.txt
          18 File(s)        115,382 bytes
          1 Dir(s)  200,809,840,640 bytes free

C:\compiler design>set path=C:\Program Files (x86)\GnuWin32\bin

C:\compiler design>flex labexp21.l

C:\compiler design>set path=C:\Program Files\CodeBlocks\MinGW\bin

C:\compiler design>gcc lex.yy.c

C:\compiler design>a.exe
G:\lex>flex count_line.l
233
^Z
Number of characters = 29
Number of words = 3
Number of lines = 2

C:\compiler design>
```

Lab exp 22 :

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex constants.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
Enter File Name : sample.c
314 is a constant
30 is a constant
Number of Constants : 2

```

Lab exp23:

```

Number of Constants : 2

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex macros.l
flex: can't open macros.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex macros.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe sample.c
Number of macros defined = 1
Number of header files included = 2

```

Lab exp24:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flrex html.l
'flrex' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex html.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
Enter File Name : sample.html
<html>
<body>
<h1>
</h1>
<p>
</p>
</body>
</html>

```

Lab exp 24:

```

Number of html tags: 8
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex lines.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
^Z

```

Lab exp25:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex email.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
mike89@gmail.com

^Z
Accepted

```

Lab exp26:

```
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex substrin.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
abcjyjkabc
ABCjyjkABC
```

Lab exp27:

```
C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex token.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe sample.c
define is a identifier
P is a identifier
314 is a number
include is a keyword
stdio.h is a standard library
include is a keyword
conio.h is a standard library
void is a keyword
main is a keyword
int is a keyword
a is a identifier
b is a identifier
c is a identifier
30 is a number
printf is a keyword
"hello" is a string literal
number of identifiers = 5
number of keywords = 6
```

Lab exp28:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex token.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe sample.c
define is a identifier
P is a identifier
314 is a number
include is a keyword
stdio.h is a standard library
include is a keyword
conio.h is a standard library
void is a keyword
main is a keyword
int is a keyword
a is a identifier
b is a identifier
c is a identifier
30 is a number
printf is a keyword
"hello" is a string literal
number of identifiers = 5
number of keywords = 6

```

Lab exp29:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex nw.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
a contains 10 and b have 2

a is WORD
contains is WORD
10 is NUMBER
and is WORD
b is WORD
have is WORD
2 is NUMBER

```

Lab exp30:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex pn.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
10
positive number = 10

-89
negative number = -89

```

Lab exp31:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex url.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe

Enter URL : https:\\www.sse.in

URL Invalid

https://www.sse.in

URL Valid

```

Lab exp32:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex dob.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
26/07/1995

```

Lab exp33:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex digit.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
10

Valid digit

e

Invalid digit

```

Lab exp34:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex math.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
20+63
The answer is = 83.000000
20*89
The answer is = 1780.000000|

```

Lab exp35:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lexical.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
enter the string : a=b+c*10
  identifiers : a b c
  constants  : 10
  operators  :

```

Lab exp36:

```

number of comments are = 2
#include<stdio.h>
int main()
{
int a,b,c;
printf("enter two numbers");
scanf("%d %d",&a,&b);
c=a+b;
printf("sum is %d",c);
return 0;
}

```

Lab exp38:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex token.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files
s (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe sample.c
define is a identifier
P is a identifier
314 is a number
include is a keyword
stdio.h is a standard library
include is a keyword
conio.h is a standard library
void is a keyword
main is a keyword
int is a keyword
a is a identifier
b is a identifier
c is a identifier
30 is a number
printf is a keyword
"hello" is a string literal
number of identifiers = 5
number of keywords = 6

```

Lab exp39:


```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex substrin.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
abcjyjkabc
ABCjyjkABC

```

Lab exp40:

```

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\GnuWin32\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>flex vowel.l

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>set path=C:\Program Files (x86)\CodeBlocks\MinGW\bin

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>gcc lex.yy.c

C:\Users\bmahe\OneDrive\Desktop\compiler design lab>a.exe
Enter the string of vowels and consonants:my name is maheshbabu

^Z
Number of vowels are: 7
Number of consonants are: 11

```